DESIGN AND FABRICATION OF MAGNETIC ANGLE ADJUSTMENT FOR WELDING

MEB1204 - MANUFACTURING TECHNOLOGY- II

Submitted by

DEENADHAYALAN S 8115U24ME010

in partial fulfilment for the award of the degree

of

BACHELOR OF ENGINEERING

In

MECHANICAL ENGINEERING

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

SAMAYAPURAM, TRICHY

ANNA UNIVERSITY

CHENNAI 600 025

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Mr. S. RAGHUVARAN., M.E., (Ph.D.)

Department of Mechanical Engineering

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

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BONAFIDE CERTIFICATE

Certified that this project report titled "**DESIGN AND FABRICATION OF MAGNETIC ANGLE ADJUSTMENT FOR WELDING**" is the bonafide work of **" DEENADHAYALN S (8115U24ME010) "**who carried out the work under my supervision.

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DECLARATION BY THE CANDIDATE

I declare that to the best of my knowledge the work reported here in has been composed solely by myself and that it has not been in whole or in part in any previous application for a degree.

Submitted for the project Viva- Voce held at K. Ramakrishnan College of Engineering on _____

SIGNATURE OF THE CANDIDATE

ACKNOWLEDGEMENT

We thank the Almighty God, for showing abundance of grace, without his blessings it would not have been possible for us to complete our project.

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We would like to thank our parents and friends for their constant support and encouragement throughout this project.

INSTITUTE VISION AND MISSION

VISION

- ❖ To achieve a prominent position among the top technical institutions

MISSION

- ❖ To bestow standard technical education par excellence through state-of-the-art infrastructure, competent faculty and high ethical standards.
- ❖ To nurture research and entrepreneurial skills among students in cutting edge technologies.
- ❖ To provide education for developing high-quality professionals to transform the society.

DEPARTMENT VISION AND MISSION

VISION

To provide eminent mechanical engineering to the industries and society through quality education.

MISSION

M1. To impart quality professional education through state-of-the-art infrastructures.

M2. To motivate students and faculty to undertake research and development activities in the thrust areas of mechanical engineering.

M3. To carry out collaborative projects with industries and academics

The project titled “Design and Fabrication of Magnetic Angle Adjustment for Welding Process” focuses on developing a low-cost, durable, and efficient fixture that helps welders maintain accurate and consistent angles while performing welding operations. Traditional welding methods rely heavily on manual clamping and angle judgment, which often leads to irregular bead formation, increased setup time, and operator fatigue. The proposed magnetic fixture overcomes these limitations by integrating strong permanent magnets, a mechanically adjustable arm, and a manual locking mechanism that ensures stable angle positioning throughout the welding process. The design includes a mild-steel base plate for structural strength, an adjustable tilting arm for angle variation, and ferrite or neodymium magnets to provide firm holding without the need for electrical power. Materials are selected based on strength, heat resistance, cost efficiency, and manufacturability to ensure the fixture performs effectively in real workshop conditions. Experimental testing showed that the magnetic angle adjustment tool improves welding precision, reduces setup time, enhances safety, and offers better ease of handling compared to conventional clamping methods. The system is highly suitable for small fabrication units, educational workshops, and industrial applications where repeated angle welding is required. Overall, the project contributes to improved welding quality, reduced human error, and enhanced productivity by applying simple mechanical principles and magnetic technology.

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CHAPTER 1

(INTRODUCTION)

Welding plays a vital and irreplaceable role in the modern manufacturing, fabrication, and construction industries. It serves as the fundamental process used to permanently join metal components to create structures, machine frames, pipelines, automotive parts, support brackets, industrial equipment, and countless other products that form the backbone of today's engineering systems. The efficiency, safety, and structural integrity of welded components strongly depend on the accuracy with which the pieces are positioned before and during welding. One of the most critical aspects of welding preparation is ensuring the correct and consistent angle between the workpieces. A slight deviation in angle can drastically affect weld penetration, bead shape, joint strength, and overall quality of the welded assembly. As industries continue to aim for higher precision, lower production cost, and improved consistency, the importance of proper workpiece alignment becomes even more significant.

Traditionally, welders depend on manual clamping tools, improvised supports, corner jigs, and sometimes even hand-held arrangements to position the metal pieces before welding. While experienced welders may achieve acceptable accuracy using manual techniques, these methods are inherently inconsistent, time-consuming, and dependent on the welder's skill level. Manual clamping often leads to challenges such as inaccurate angular positioning, movement during welding due to heat distortion, and difficulty in holding awkwardly shaped components. In small-scale workshops, where precise fixtures and advanced welding positioners are rarely available due to high cost, the issue of maintaining correct angles becomes even more problematic. As a result, weld quality becomes inconsistent, and defects such as lack of fusion, misalignment, porosity, undercut,

and uneven bead formation may occur. These issues lead to additional rework, wastage of materials, higher labor time, and reduced overall productivity.

To address these common challenges observed in conventional welding setups, there is a growing demand for a simple, reliable, cost-effective, and easy-to-use solution that ensures accurate angle maintenance without requiring complex equipment. This necessity has led to the development of the Magnetic Angle Adjustment Tool, a mechanical fixture that uses the power of permanent magnets combined with an adjustable arm to hold workpieces at precise angles during welding. The intention behind this device is to eliminate the shortcomings of manual clamping techniques and provide welders with a standardized fixture that ensures repeatability and accuracy irrespective of operator skill. By integrating strong magnets and mechanical locking arrangements, the tool offers a practical method to hold metal components in place during welding operations without relying on electricity or external power sources.

The Magnetic Angle Adjustment Tool is designed as a compact, lightweight, and user-friendly fixture incorporating key components such as a sturdy mild-steel base plate, an adjustable tilting arm, high-strength ferrite or neodymium magnets, and a mechanical locking mechanism. The combination of these components ensures that workpieces can be held securely at commonly used angles such as 30°, 45°, 60°, 75°, 90°, or any other angle as required. The magnets used in the tool provide firm holding force, preventing the workpiece from sliding or shifting during welding. The adjustable arm allows the welder to set the desired angle accurately and lock it in place, ensuring consistent results throughout the operation. The entire assembly is designed to withstand welding heat, spatter, and workshop environmental conditions, making it highly suitable for industrial and educational use.

A major advantage of this tool is that it requires no electrical supply, unlike conventional automated or semi-automated welding positioners. This makes the magnetic fixture extremely economical and accessible to small welding shops, fabrication units, and college workshops where budget constraints limit the availability of advanced equipment. The project emphasizes the use of basic workshop processes such as cutting, drilling, welding, grinding, and assembly to fabricate the fixture. These fabrication steps provide students with valuable hands-on experience in manufacturing technologies, enhancing their understanding of machining, mechanical design, and materials engineering. By designing and building the fixture themselves, students also develop practical knowledge of fixture design principles, load-bearing considerations, and the importance of selecting suitable materials for mechanical systems.

The primary purpose of this project is to create a tool that improves the accuracy, stability, and safety of welding operations. By maintaining the required angle precisely throughout the welding process, the magnetic fixture helps reduce human error and leads to improved weld quality. This also reduces the physical strain on welders, as they no longer need to manually hold or repeatedly adjust workpieces. The tool significantly speeds up setup time, ensuring higher productivity and efficiency in welding environments. Even inexperienced welders can achieve consistent results when using the magnetic fixture, thereby enhancing the reliability of welding operations in educational settings and small-scale industries.

Testing of the fabricated magnetic fixture shows that it provides excellent holding strength, stable angle adjustment, and repeatable welding alignment. It ensures consistent weld bead formation and reduces distortion caused by poor clamping in manual setups. The simplicity of the design, combined with its effectiveness, demonstrates how basic engineering principles—such as permanent magnetism

and mechanical leverage—can be used to solve everyday industrial problems. Furthermore, because the fixture eliminates the need for costly angle gauges, clamps, or automated positioners, it offers an affordable alternative for workshops aiming to improve welding accuracy without making large financial investments.

The Magnetic Angle Adjustment for Welding Process project is a practical and innovative solution that addresses a critical issue faced by welders across different industries. By combining fundamental mechanical concepts with magnetic holding technology, the project successfully presents a functional, durable, and cost-effective welding fixture that enhances precision, reduces human effort, and promotes safe working conditions. It aligns with academic goals by offering students real-world engineering experience and fosters creativity, problem-solving, and technical competence. This magnetic fixture is especially beneficial for diploma and degree students, small fabrication shops, and engineering institutions seeking efficient and economical methods to improve welding accuracy and consistency. The project ultimately demonstrates how simple mechanical systems can significantly improve manufacturing quality without relying on complex or expensive technologies.

CHAPTER 2

LITERATURE SURVEY

1. L. Zhang et al. (2018) – Design and Analysis of Permanent Magnet Fixture for Complex-Shaped Components

L. Zhang and his team (2018) investigated advanced fixturing solutions using permanent magnets for holding complex-shaped metallic components during machining and welding. The study focused on optimizing magnetic flux distribution and calculating magnetic holding force under different loading conditions. Through simulation and experimental testing, the authors demonstrated that properly designed magnet fixtures can offer significantly stronger gripping force with reduced weight and size. Their research showed that magnetic fixtures eliminate slippage and improve precision during assembly or welding operations. However, the system required careful magnet placement and shape optimization, making it more complex for small workshops. (Source: Journal of Manufacturing Engineering, 2018.)

2. M. V. Kulkarni et al. (2020) – Review on Jigs and Fixtures Design and Development in Welding Automation

Kulkarni and colleagues (2020) provided a comprehensive review of welding jigs and fixtures, highlighting the increasing demand for adjustable, modular, and quick-setup fixturing systems. The paper discussed limitations of conventional clamping devices and emphasized the need for solutions that support precise angular positioning and fast reconfiguration. The authors pointed out that magnetic fixtures offer superior speed and operator convenience, making them ideal for high-mix, low-volume welding operations. However, they noted that magnetic force may reduce under high heat or contamination. (Source: International Journal of Production Research, 2020.)

3. S. Kumar & A. Sharma (2015) – Development of an Adjustable Angle Welding Clamp Using Mechanical Locking

Kumar and Sharma (2015) developed a purely mechanical adjustable welding clamp that enables welders to set multiple angles using a gear-based locking mechanism. Their design improved repeatability and reduced setup time for angular joints, enhancing weld accuracy. The study highlighted that rigid locking prevents angular drift during welding. However, the authors acknowledged that mechanical clamps require frequent tightening and may wear out under high torque loads — unlike magnetic fixtures, which maintain force consistently without user effort. (Source: Indian Journal of Mechanical Engineering, 2015.)

4. J. Wang (2019) – Magnetic Force Calculation and Application in Adjustable Welding Devices

Wang (2019) proposed mathematical models to compute required magnetic holding force for various welding loads and angles. The research analyzed the relationship between material thickness, magnetic grade, air gap, and shear force during welding. The findings helped define safe working limits for ferrite and neodymium magnets in welding fixtures. The study concluded that permanent magnets are practical and highly efficient for adjustable angle welding tools. However, accuracy depends on correct magnet selection and mounting structure. (Source: International Journal of Applied Physics, 2019.)

5. P. R. Jones (2021) – Use of Rare-Earth Magnets in Industrial Tooling and Fixturing

Jones (2021) investigated the properties of rare-earth magnets, especially neodymium magnets, and evaluated their suitability for industrial fixturing

applications. The study emphasized their excellent strength-to-weight ratio and ability to maintain consistent holding force even in compact designs. This makes them ideal for lightweight, portable welding fixtures. The authors also discussed demagnetization factors, temperature limitations, and coating requirements. The research reinforced the advantage of using neodymium magnets in adjustable welding angle tools. (Source: Materials and Design Journal, 2021.)

6. H. S. Lim (2017) – Ergonomics and Efficiency in Welding Through Innovative Jig Design

Lim (2017) explored how ergonomically engineered welding fixtures reduce welder fatigue, improve posture, and increase productivity. The study highlighted that magnetic clamps significantly speed up part positioning compared to manual clamps and reduce the physical strain caused by repetitive tightening. The research demonstrated productivity gains in small and medium-scale welding shops using magnetic jigs. The limitation was the reduced performance of magnets on rusty or uneven metal surfaces. (Source: Journal of Ergonomics in Engineering, 2017.)

7. G. K. Singh (2016) – Finite Element Analysis (FEA) of Magnetic Clamping Devices

Singh (2016) used FEA to analyze magnetic fixturing systems exposed to external welding forces such as shear load, heat, and vibrations. The study predicted holding capacity, stress distribution, and potential failure points in magnetic clamps. Results confirmed that permanent magnets maintain reliable grip if supported by proper mechanical housing. The research also highlighted that

magnet performance decreases with heat, requiring proper insulation or air gaps. (Source: International Journal of Simulation and Modelling, 2016.)

8. A. B. Chen (2014) – Design and Prototyping of Modular Welding Fixture Systems

Chen (2014) presented a modular fixturing system combining permanent magnets and reconfigurable elements that can be rearranged for different joint types and welding angles. This system significantly reduced fixture setup time and allowed precise, repeatable positioning. The study concluded that modularity is crucial for low-cost workshops needing flexible tools. However, the design required higher manufacturing precision, increasing initial fabrication cost. (Source: Journal of Manufacturing Automation, 2014.)

9. T. Mori (2018) – Evaluation of Angle Accuracy and Stability in Welding Jigs

Mori (2018) conducted experimental studies on angular stability of welding jigs under high heat conditions. The paper compared mechanical clamps and magnetic fixtures, finding that magnetic jigs provide faster setup but require careful structural reinforcement to maintain angle stability. The research analyzed tolerance variations and showed that adjustable angle fixtures improve overall weld quality for repetitive tasks. (Source: International Welding Engineering Journal, 2018.)

10. K. L. Wu (2022) – Fabrication Techniques for Precision Metal Components in Tool Design

Wu (2022) explored various modern manufacturing methods such as CNC machining, laser cutting, and 3D printing for producing precise jig and fixture

components. The study discussed material selection for magnetic fixtures, focusing on mild steel, aluminum, and alloy steels. Results highlighted that accurate fabrication improves angle repeatability and enhances magnetic coupling efficiency. (Source: Journal of Tool Engineering, 2022.)

11. D. Pathak (2020) – Effect of Electrode Angle and Welding Current on Tensile Strength of SMAW Welds

Pathak (2020) experimentally studied how electrode angles between 65° and 85° influence tensile strength in SMAW welding. The study found that larger electrode angles improved penetration and tensile strength, emphasizing the importance of correct angle positioning during welding. This supports the need for adjustable angle fixtures. (Source**:** Welding Research Bulletin, 2020.)

12. H. R. Ghazvinloo (2019) – Effect of Electrode-to-Work Distance and Angle on Weld Geometry

Ghazvinloo (2019) analyzed how changing the electrode angle from 65° to 75° improves penetration depth, while also noting that filler diameter affects weld bead shape. The study underlines the significance of precise angular control, making adjustable magnetic fixtures valuable. (Source: Journal of Materials Processing Technology, 2019.)

13. Muhammad Abid (2020) – Influence of Welding Torch Inclination on Weld Bead Formation

Abid (2020) studied welding torch inclination angles (30°, 60°, 90°, 120°) and found strong effects on weld pool shape, bead width, and heat distribution. The research demonstrated the necessity of maintaining consistent torch and

workpiece angles — reinforcing the usefulness of angle adjustment tools. (Source: International Journal of Advanced Welding Technology, 2020.)

14. M. Abid (2013) – Effect of Torch Angle in GTAW Processes

This 2013 study used 3D simulation to analyze how torch angles of 90° and 70° influence arc properties like current density and heat flux. Variations caused changes in weld penetration and bead profile. Accurate angle holding tools were recommended for precision welding. (Source: Journal of Thermal Engineering, 2013.)

15. H. Hafni (2019) – Electrode Angle Influence in Forward Welding on Fusion Depth

Hafni (2019) evaluated multiple angles (70°, 75°, 80°, 85°, 90°) in SMAW forward welding and concluded that penetration depth and weld strength depend heavily on electrode angle. This further justifies the development of adjustable magnetic angle welding fixtures. (Source: International Welding Science Journal, 2019.)

CHAPTER-3

PROBLEM IDENTIFICATION

1. Maintaining accurate and consistent welding angles manually is difficult, leading to improper joint alignment.

2. Traditional welding clamps often slip or loosen, causing angle variations during welding.

3. Manual positioning requires repeated adjustments, increasing welding setup time.

4. Workpiece misalignment results in uneven weld beads and weak joint strength.

5. Holding the workpieces at specific angles such as 30°, 45°, or 90° without fixtures is challenging.

6. Welders must rely heavily on experience and visual judgment, increasing human error.

7. Small workshops lack access to precise and costly welding angle fixtures.

8. Without proper angle control, heat distortion causes poor dimensional accuracy.

9. Manual clamping methods fatigue the welder and reduce productivity in long welding tasks.

10. Low-cost traditional jigs do not provide adjustable angle flexibility for various weld operations.

11. Absence of reliable holding systems leads to frequent rework and wastage of materials.

12. Inconsistent manual positioning reduces repeatability and uniformity in welded components.

13. Welders face difficulty keeping the workpiece stable while handling the welding torch simultaneously.

14. A simple, low-cost, and efficient magnetic fixture is needed to achieve stable angle holding and improve weld accuracy

In most fabrication and manufacturing industries, welding is still performed using manual positioning methods such as traditional clamps, handheld supports, and improvised fixtures. These manual methods often lead to several performance, safety, and quality issues during welding. Without proper angular control, workpieces frequently shift or misalign, resulting in inconsistent weld beads, weak joints, and poor surface finish. The accuracy of the weld is highly dependent on the welder's experience and steadiness, making it difficult to maintain uniform standards across repetitive tasks or multiple components. In addition, manually holding or adjusting angles increases human effort, causes fatigue, and reduces productivity, especially during long welding cycles. Maintaining precise angles such as 45°, 60°, or 90° manually is challenging, leading to distortion, rework, and increased production time. Small workshops often lack access to precise welding fixtures or adjustable positioners due to high cost, forcing welders to rely on skill-based judgment. Therefore, there is a strong need to identify and resolve these issues by developing a simple, low-cost, durable, and user-friendly magnetic angle adjustment tool that enables accurate positioning, reduces manual workload, improves welding consistency, and enhances the overall quality and productivity of welding operations.

CHAPTER 4

OBJECTIVE AND SCOPE OF THE PROJECT

The primary objective of this project is to design and fabricate a simple, low-cost, and durable magnetic angle adjustment fixture that improves the accuracy, stability, and efficiency of welding operations. Traditional welding methods rely heavily on manual clamping and the welder's skill to maintain correct angles,

which often leads to misalignment, poor weld quality, and additional rework. The project aims to eliminate these limitations by developing a fixture that uses permanent magnetic holding force combined with a mechanical angle-locking system, ensuring that metal workpieces remain secure and properly aligned throughout the welding process.

A key objective is to create a mechanically robust and user-friendly device using basic materials such as mild steel, ferrite or neodymium magnets, and simple locking components. The fixture must allow welders to set and hold different angles—such as 30°, 45°, 60°, 90°, and intermediate angles—without slipping or readjustment. Unlike commercial adjustable fixtures that are costly, the project seeks to create an economical solution suitable for small workshops, fabrication units, and educational laboratories.

Another important objective is to enhance safety and reduce operator fatigue by minimizing manual handling of workpieces during welding. By securing the components using magnetic force, the fixture ensures a stable working environment and reduces the chance of workpiece shifting or accidental burns during welding.

The project also aims to provide practical learning opportunities for students by involving various fabrication processes, including cutting, drilling, filing, grinding, and welding. Through the construction of the model, students gain experience in fixture design, material selection, angle measurement, and assembly practices—strengthening their understanding of manufacturing technology.

A further objective is to ensure repeatability and consistency in angle holding, which helps standardize welding operations and reduce quality variations.

Testing of the fabricated model focuses on evaluating magnetic holding strength, angle accuracy, stability under load, and ease of operation.

Overall, the objective of the project is to deliver a functional, economical, and efficient magnetic angle adjustment tool that improves weld precision, supports safer welding practices, minimizes errors, and serves as an educational model for engineering and diploma students.

SCOPE OF THE PROJECT

The scope of this project includes the design, fabrication, testing, and evaluation of a magnetic angle adjustment fixture specifically created for welding applications. The project covers all stages from concept development to prototype validation but remains focused on creating a workshop-ready fixture rather than developing advanced automated or commercial-grade systems.

The project begins with conceptual design, where different geometric arrangements of the magnetic base, adjustable arm, and locking mechanism are evaluated. This includes selecting appropriate magnet types (ferrite or neodymium), analyzing holding force requirements, and determining suitable angles for welding operations. Simple hand sketches and CAD models are prepared to finalize the overall structure.

Next, the project covers detailed engineering, which includes selecting cost-effective and durable materials such as mild steel for the frame, stainless steel or coated bolts for locking mechanisms, and magnets chosen based on temperature resistance and strength. Consideration is given to keeping the design compact, lightweight, and suitable for repeated workshop use.

The scope further includes fabrication and assembly, using basic manufacturing processes available in a college workshop. This involves cutting MS plates and bars, drilling pivot holes, welding joints, grinding surfaces for smoothness, and securely fixing the magnets and locking components. The assembly stage ensures that all moving parts function smoothly and the desired angles can be set accurately.

A major part of the scope involves testing and performance evaluation. The fixture is tested for magnetic holding strength, angle accuracy, stability during welding, resistance to heat, and ease of use. These tests determine whether the tool meets the intended objectives of improving weld quality and reducing human error.

Additionally, the scope includes preparing technical documentation, such as 2D drawings, 3D models, bill of materials (BOM), fabrication procedures, and performance results. This helps in verifying that the fixture complies with functional requirements and can be reproduced or improved in future projects.

The scope is limited to creating a prototype model for educational and workshop usage. It does not include mass-production methods, advanced automation, robotic welding integration, or commercial certification processes. However, the design provides a foundation for future enhancements such as digital angle indication or modular attachments.

In summary, the scope of the project is to develop a practical and economical magnetic welding fixture through a structured engineering approach involving design, material selection, fabrication, analysis, and validation—offering a reliable tool that enhances welding precision and provides valuable learning for mechanical engineering students.

CHAPTER 5

MATERIALS AND METHODS

COMPONENT	DETAILS
	Electric spray gun or solenoid-controlled gun
Chain & Chain Brackets	Bicycle chain type or industrial timing chain
Sprockets	For stepper motor shaft and frame
Steel Frame	Square tube frame for horizontal rail
4 Tyres / Wheels	Small PU wheels or steel wheels for smooth horizontal movement
Gun Lifting Mechanism	Small DC gear motor or linear actuator for gun up–down
Guide Rails	Steel pipes or rods for smooth horizontal travel
Bearing Blocks	For low friction movement
Mounting Plate / Motor Brackets	For fixing stepper motor & spray gun assembly

The automatic paint sprayer system was constructed using an Arduino microcontroller as the primary control unit, a stepper motor for vertical gun movement, and a DC motor or conveyor mechanism to enable horizontal motion of the tyre. Limit switches were installed at the upper and lower ends of the gun's travel path to provide positional feedback and prevent over-travel. A relay module was incorporated to control the paint sprayer activation and motor power switching. The spray gun was mounted on a guided rail system that allows smooth upward and downward motion. During operation, the tyre moves horizontally on a conveyor or motor-driven platform while the Arduino executes the pre-

programmed code stored in its memory. As the gun reaches either limit switch, the signal sent to the microcontroller triggers a direction change in the stepper motor, causing the spray gun to reverse its vertical motion. Simultaneously, the relay board activates the paint sprayer so that continuous spraying occurs throughout the upward and downward strokes. The coordinated movement of the sprayer and tyre ensures uniform coating, while feedback from the limit switches and controlled motor actuation maintains accuracy, repeatability, and safety during the entire process.

The mechanical structure of the system consists of a rail-mounted paint gun and a horizontally movable tyre platform. The vertical motion of the paint gun is guided by a stepper motor attached to a lead screw or belt mechanism, ensuring precise and smooth movement. Limit switches are positioned at the upper and lower extremes of the vertical path to act as safety stops, preventing mechanical overrun and potential damage to the equipment. The tyre is mounted on a motor-driven horizontal conveyor that moves at a controlled speed to synchronize with the vertical spraying pattern. Proper alignment of the spray gun and tyre is maintained through adjustable mounts and brackets, allowing uniform coverage during the painting process.

The electrical control system integrates an Arduino microcontroller, a relay board, and stepper motor drivers. The Arduino receives signals from the limit switches to detect the end positions of the vertical movement. Based on this feedback, it controls the stepper motor direction and speed to achieve smooth reciprocating motion. The relay board acts as a switch to turn the paint sprayer on and off, coordinating spraying with the gun's vertical strokes. Power is supplied through regulated sources to ensure stable operation of motors, relays, and the microcontroller. Wiring is organized with proper insulation to prevent short circuits and ensure safety during operation.

The Arduino microcontroller is programmed with pre-defined instructions to control the stepper motor, relay, and timing of the painting process. The code continuously monitors the status of the limit switches, triggering a change in vertical direction when the gun reaches the top or bottom positions. The horizontal movement of the tyre is synchronized with the gun's motion to maintain consistent paint coverage. Delays and speed controls are included in the program to optimize spray thickness and prevent over-application. The system can be calibrated by adjusting parameters in the code, such as stepper motor steps, delay intervals, and conveyor speed, to suit different tyre sizes and paint viscosities.

During operation, the system is first initialized by positioning the paint gun at a starting point above the tyre. The Arduino executes the program, activating the stepper motor to move the gun vertically while simultaneously running the conveyor to move the tyre horizontally. When the gun touches the limit switches, the motor reverses direction, ensuring continuous reciprocating motion. The relay board activates the paint sprayer only during the vertical strokes, allowing uniform coating of the tyre surface. This automated process reduces manual labor, increases efficiency, and ensures a consistent paint layer. Periodic maintenance of mechanical and electrical components is recommended to maintain accuracy and prolong system life.

CHAPTER 6

SELECTION OF MATERIALS

Material Selection for Magnetic Angle Adjustment Welding Tool

The magnetic angle adjustment welding tool is constructed using a combination of carefully chosen materials to ensure durability, precision, and reliability during welding operations. Each component was selected based on factors such as strength, corrosion resistance, adjustability, and suitability for high-temperature working environments commonly encountered in welding applications.

- **Neodymium Magnets**

  The holding power of the device is provided by neodymium magnets, chosen for their exceptionally high magnetic strength and compact size. These magnets ensure that the tool can securely attach to metal surfaces, allowing welders to maintain stable positioning while working at various angles. Neodymium magnets are significantly stronger than regular ferrite magnets, making them ideal for supporting heavy metal workpieces without slipping during welding. Their coating adds protection against rust, heat exposure, and surface damage, improving reliability and lifespan in industrial settings.

- **Bolt with Hook (Connected to the Magnet)**

  Bolts with hooks are attached to the magnets to allow simple and adjustable connection points for securing the device. These hooks make it easy to hang or position metal pieces or clamps during alignment. The hooks are manufactured from steel to ensure they can withstand repeated load, tension, and physical handling without bending or deformation. Their threaded connection allows the magnets to be fastened firmly to the tool, maintaining consistent placement and stability during use.

- **Angle Scale**

  An angle measuring scale is installed at the joint between the two movable bars. The scale provides precise angular adjustments, allowing welders to repeat specific angles with high accuracy. This component plays a critical role in ensuring consistent and repeatable welding alignment without relying solely on manual measurement. The angle markings are designed to remain visible under frequent handling, metal dust, and high-temperature conditions, improving usability during fabrication and welding tasks.

- **Stainless Steel Square Bar**

  The primary structure of the tool is made from a stainless steel square bar. Stainless steel was selected for its high tensile strength, rigidity, and corrosion resistance. This ensures that the tool can withstand mechanical stresses without bending, warping, or rusting. Stainless steel also provides a smooth, durable surface finish, making the tool suitable for workshop and industrial environments where accuracy and long-term reliability are essential. Its resistance to sparks and heat exposure makes it ideal for welding work.

- **Steel Pipe (Internal Support For Magnet)**

  A steel pipe is placed inside the square bar to provide additional structural support and allow smooth rotation of the adjustable arms. Steel was chosen

because of its strength, durability, and ability to maintain alignment under mechanical load. The pipe ensures stable movement between the two bars while preventing wear on the external structure. This design allows controlled rotation and precise angle adjustment, making the tool functional and long-lasting.

- **Steel Block**

At the center joint, a circular steel component is placed to connect the two bars and house the angle scale and rotational mechanism. Circular steel was selected because it provides uniform strength, stability, and wear resistance, ensuring precise rotational movement. Its round geometry ensures consistent load distribution, reducing mechanical fatigue and improving alignment accuracy over time.

- **Tightening Knob**

The tightening knob is made from plastic with a stainless steel threaded bolt, selected to provide both comfort and strength during use. The plastic handle offers a lightweight, heat-resistant, and non-conductive surface, making it safe to grip even when welding conditions involve high temperatures. Its ergonomic shape allows easy tightening and loosening, even while wearing gloves. The stainless steel thread provides durability, corrosion resistance, and secure fastening, ensuring the adjustable bars remain firmly locked in place during operation.

CHAPTER 7

WORKING PRINCIPLE

The magnetic angle adjustment welding tool operates by allowing precise rotation and locking of the connected metal bars to achieve the required welding angle. The central circular steel joint houses the angle scale, which

enables accurate measurement during adjustment. Neodymium magnets positioned on the tool securely attach to metal workpieces, ensuring stable positioning without manual holding. The adjustable bars can be rotated to the desired angle, and once aligned, the stainless-steel threaded tightening knob locks the mechanism firmly in place to prevent movement during welding. The internal steel pipe provides smooth rotation and structural support to the joint, while the hooks assist in securing additional fixtures if required. Through the combined function of the rotating mechanism, strong magnetic holding, and locking knob, the tool allows welders to maintain consistent angles accurately and hands-free, improving efficiency, precision, and safety during welding tasks.

BRIEF EXPLANATION ABOUT PRINCIPLE

The magnetic angle adjustment welding tool operates on the principle of controlled mechanical positioning using magnetic holding force and adjustable angular movement. The system uses strong neodymium magnets to firmly attach the tool to ferrous metal surfaces, eliminating the need for manual support while welding. These magnets hold the workpiece or welding reference point securely in place, ensuring stability throughout the welding process.

The angle adjustment is achieved through a rotating mechanism located at the center of the tool, which allows the two metal bars to move relative to each other. A precision angle scale is integrated into the joint, enabling accurate alignment to common welding angles such as 45°, 90°, or customized positions based on requirement. Once the desired angle is set, a tightening

knob with a threaded steel fastener locks the adjustable bars, preventing movement and maintaining the final angle during welding.

The internal steel pipe structure ensures smooth and stable rotation between the bars, while the circular steel centre distributes mechanical loads evenly, preventing deformation under repeated use. Hook bolts attached to the magnets allow additional support or clamping functions when handling larger or irregular components.

This coordinated operation between magnetic attachment, adjustable angular movement, and mechanical locking ensures that the tool provides hands-free, repeatable, and precise alignment during welding. The principle allows welders to maintain consistent angles, reduce setup time, improve accuracy, and enhance safety by avoiding manual holding or measurement during welding tasks.

CHAPTER 8

PROCESS INVOLVED IN FABRICATION

CUTTING

The fabrication process begins with cutting raw materials such as metal rods, sheets, and support brackets to the required dimensions. Tools like hacksaws, angle grinders, or CNC cutting machines are used depending on the accuracy needed. Proper marking and measurement are completed before cutting to ensure all parts match the desired specifications and fit correctly during later assembly.

WELDING / JOINING

After cutting, the components are joined together using welding or bolting methods. Welding is mainly used where a strong and permanent connection is required, especially for the main supporting frame. In areas where adjustments or dismantling may be needed, nuts and bolts are used instead. Alignment during this step is important to ensure smooth movement and precision in the final mechanism.

DRILLING

Drilling is carried out to create holes for mounting magnets, pivot joints, fasteners, and adjustment mechanisms. A hand drill or pillar drill may be used depending on the accuracy required. After drilling, burrs and sharp edges are removed to ensure proper fitting and smooth movement of mechanical components.

GRINDING AND FINISHING

Once drilling and joining are completed, grinding is performed to remove rough welds, sharp edges, and unwanted metal projections. Finishing improves both the appearance and safety of the system while reducing friction between moving parts. This ensures that the magnetic angle adjustment mechanism operates smoothly and reliably.

ASSEMBLY

In the final stage, all fabricated components are assembled, including the frame, magnetic mounts, pivot mechanism, locking system, and adjustment controls. Each part is fitted accurately and aligned to ensure proper functionality. The system is then tested to confirm that the magnetic angle adjustment works smoothly and securely at different positions.

The fabrication of the Magnetic Angle Adjustment Tool for welding involves a series of systematic steps to ensure accuracy, durability, and stable operation during welding tasks. Initially, raw materials such as mild steel plates, square tubes, and flat bars are cut to the required dimensions using cutting machines, hacksaws, or angle grinders to achieve precise structural measurements. Once the materials are cut, the base frame and adjustable arm components are joined using appropriate welding methods to create a rigid and stable structure capable of withstanding magnetic loads and welding heat. Drilling operations are performed to create accurately positioned holes for mounting the tilting arm, locking pin mechanisms, adjustment slots, and magnet housings, ensuring smooth rotation and reliable angle locking.

After cutting, welding, and drilling, all fabricated parts undergo grinding and surface finishing to remove burrs, ensure smooth edges, and improve the fitting of moving components. This finishing process also enhances the safety and handling comfort of the fixture. Following the finishing stage, the permanent magnets—either ferrite or neodymium—are carefully installed into their housings or brackets, ensuring firm seating and maximum holding force. The adjustable arm assembly, locking mechanism, and magnetic housing are then fitted onto the main frame with precise alignment to allow smooth and repeatable angle adjustments.

Finally, all components are assembled and checked for movement, locking stability, and holding strength. The completed fixture is tested for different angular positions such as 30°, 45°, 60°, and 90° to evaluate its consistency, magnetic grip, and ease of adjustment during welding. This structured fabrication process ensures that the Magnetic Angle Adjustment Tool is robust, user-friendly, and capable of providing reliable angle support during welding operations.

CHAPTER 12

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